

A Literature-Implied Sharpening of the ℓ_1^n Helly Radius

Math discovery packet

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1 Status

This packet records a *literature-implied* correction/sharpening, not a new original lower-bound proof. The source paper is

G. Ivanov, *On Banach Spaces with the Helly Approximation Property*, arXiv:2603.22743.

The supporting paper is

V. Reis and T. Rothvoss, *Approximate Carathéodory bounds via Discrepancy Theory*, arXiv:2207.03614.

The source paper obtains $r_k(\ell_1^n) \leq C\sqrt{\log n/k}$ and then conjectures that this bound is optimal.

On the other hand, the approach from [Iva25]; on the dual side, this corresponds to replacing ℓ_∞^n by ℓ_p^n , where p and q are conjugate exponents. The same estimate also follows from Theorem 1.1, since $(\ell_1^n)^* = \ell_\infty^n$ and

$$T_2(\ell_\infty^n) \leq C\sqrt{\ln n}.$$

We conjecture that the bound $r_k(\ell_1^n) \leq C\sqrt{\frac{\ln n}{k}}$ is optimal.

On the other hand, our examples show that

2 Claim

Theorem 1. *Let $h_k(\ell_1^n)$ be the optimal radius in the no-dimensional Helly definition of arXiv:2603.22743. For $1 \leq k \leq n$,*

$$h_k(\ell_1^n) \leq C\sqrt{\frac{\log(2n/k)}{k}}$$

for a universal constant C . Consequently, the displayed $\sqrt{\log n/k}$ bound in the source paper is not uniformly optimal in k as written.

3 Supporting Literature

Reis–Rothvoss define $\text{ac}_k(P, Q)$ as the worst-case error when approximating a point of $\text{conv}(X)$, $X \subseteq P$, by the uniform average of k points of X , measured in the Q -norm. Their Corollary 5 states:

bound in [MLVW17] whenever $p \ll \log(\frac{2m}{k})$:

Corollary 5. *Let $2 \leq p \leq \infty$ and $k \in \mathbb{N}$. Then*

$$\text{ac}_k(B_p^m, B_p^m) \lesssim \sqrt{\frac{\min\{p, \log(\frac{2m}{k})\}}{k}}$$

The vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ can be found in time $O(\log^2 m) \cdot T(m, m+1) \leq O(m^{1+\omega} \log^2(m))$

Taking $p = \infty$ and $m = n$ gives

$$\text{ac}_k(B_\infty^n, B_\infty^n) \leq C_0 \sqrt{\frac{\log(2n/k)}{k}}.$$

4 Transfer to Helly

Lemma 1. *Let X be finite-dimensional. If*

$$\text{ac}_k(B_{X^*}, B_{X^*}) \leq \varepsilon,$$

then 3ε is a valid no-dimensional Helly radius for X .

Proof. Let $\mathcal{F}$ be a finite family of closed convex subsets of X such that every k -element subfamily intersects B_X . Put

$$f(x) = \max_{K \in \mathcal{F}} \text{dist}(x, K)$$

and choose a minimizer x_0 . Since every $K \in \mathcal{F}$ intersects B_X , $f(0) \leq 1$, and hence $x_0 \in 2B_X$.

Let $r = f(x_0)$. If $r = 0$ there is nothing to prove. Otherwise the standard subdifferential argument used in arXiv:2603.22743 gives active sets $K_1, \dots, K_m$ and functionals $u_1, \dots, u_m \in B_{X^*}$ with

$$0 \in \text{conv}\{u_1, \dots, u_m\}$$

and

$$\langle u_i, x - x_0 \rangle \leq -r \quad (x \in K_i).$$

By the definition of ac_k , choose indices $i_1, \dots, i_k$ such that

$$\left\| \frac{1}{k} \sum_{t=1}^k u_{i_t} \right\| \leq \varepsilon.$$

Repeated indices cause no difficulty: the distinct chosen active sets can be extended to a k -element subfamily, so the Helly hypothesis supplies

$$q \in B_X \cap \bigcap_{t=1}^k K_{i_t}.$$

Average the displayed inequalities at q :

$$r \leq - \left\langle \frac{1}{k} \sum_{t=1}^k u_{i_t}, q - x_0 \right\rangle \leq \varepsilon \|q - x_0\| \leq 3\varepsilon.$$

Thus x_0 lies within 3ε of every set in $\mathcal{F}$. □

Apply the lemma to $X = \ell_1^n$, so $X^* = \ell_\infty^n$, and use the Reis–Rothvoss corollary. This proves the claimed bound.

5 Interpretation

The source paper’s conjecture is not wrong if read as a rough statement in regimes where $\log(2n/k)$ and $\log n$ are comparable. But it cannot be correct as a uniform-in- k lower-bound assertion. For example, when $k = n$, the bound above gives

$$h_n(\ell_1^n) \leq C/\sqrt{n},$$

whereas a matching lower bound of order $\sqrt{\log n/n}$ would eventually contradict this by an unbounded factor.

6 Limitations

- This packet does not prove the matching lower bound $\Omega(\sqrt{\log(2n/k)/k})$.
- It does not settle the exact asymptotic behavior of $h_k(\ell_1^n)$.
- It records a separate-literature implication, not a new proof of the approximate-Carathéodory theorem.

7 Verification

The script `code/check_implication.py` checks that the source and supporting PDFs contain the snippets used above:

```
conda run --no-capture-output -n sandbox python code/check_implication.py
```